

$$\begin{aligned}
 0.3667 \varphi_2 - 0.2 \varphi_3 - 0.1 \varphi_4 &= 2.6667 \\
 -0.2 \varphi_2 + 0.4778 \varphi_3 - 0.1667 \varphi_4 &= -18.2222 \\
 -0.1 \varphi_2 - 0.1667 \varphi_3 + 0.3167 \varphi_4 &= 16.6667
 \end{aligned}$$

Тогда

$$G_3 = \frac{1}{R_3} = \frac{1}{2.7} = 0.366667 \text{ S}$$

$$G_2 = \frac{1}{R_2} = \frac{1}{2.1} = 0.476190 \text{ S}$$

$$G_8 = \frac{1}{R_8} = \frac{1}{20} = 0.05 \text{ S}$$

$$G_1 = \frac{1}{R_1} = \frac{1}{6} = 0.166667 \text{ S}$$

$$G_6 = \frac{1}{R_6} = \frac{1}{10} = 0.1 \text{ S}$$

Решим систему с помощью матрицы

$$\varphi = (A G_0 A^T)^{-1} (A J_0 - A G_0 B_0)$$

После расчета получим

$$\varphi_2 = 10.651 \text{ V}$$

$$\varphi_3 = -17.394 \text{ V}$$

$$\varphi_4 = 46.809 \text{ V}$$

Определим токи ветвей

$$I_1 = \frac{R_2 \varphi_2 + \varphi_3}{R_1} = \frac{0 - (10.651) + (-17.394)}{6} = -5.963 \text{ A}$$

$$I_2 = \frac{R_2 \varphi_2 + \varphi_3}{R_2} = \frac{-17.394 - (10.651)}{2.1} = -13.623 \text{ A}$$

$$I_5 = \frac{R_8 \varphi_4}{R_8} = \frac{46.809}{20} = 2.34 \text{ A}$$

$$I_1 = \frac{R_2 \varphi_2 + \varphi_3}{R_1} = \frac{-17.394 - (46.809)}{6} = -11.965 \text{ A}$$

$$I_6 = \frac{R_8 \varphi_4}{R_6} = \frac{46.809}{10} = 4.68 \text{ A}$$

$$I_0 = (\varphi_3 - \varphi_2) G_0 = (-17.394 - 10.651) \cdot 0.2 = -5.589 \text{ A}$$

$$I_0 = I_6 = 4 \text{ A}$$